

Chemistry Ch 5.6: Metallic Bonds

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Electron sea model

Dhat ke atoms apne bairooni electrons **chhor** dete hain. Ye electrons kisi **ek** atom ke nahin rehte, balke poore maddey mein **azadana ghoomte** hain.

In ko **delocalised electrons** kehte hain, aur is tasveer ko **sea of electrons** kaha jata hai.

To dhat kya hai? **Musbat ions ki jaali, jo electron ke samundar mein doobi hui hai.** Aur yehi kheench metallic bond hai.

Ek tasveer, chaar khasoosiyat

Bijli chalati hai: azad electrons charge le jate hain.

Garmi chalati hai: wohi electrons garmi bhi le jate hain.

Malleable (peet kar chaadar): peetne par atoms ki tehen **phisal** jati hain, magar electron ka samundar unhein jore rakhta hai, is liye dhat **tootti nahin, phailti** hai.

Ductile (taar): wohi wajah.

Chamak: azad electrons roshni ko wapas **parchhati** hain.

5.6.1 Sanaat mein ye khasoosiyat kyun ahem hain

Taanba (copper): bijli ki taar, kyunke wo sab se acha conductor aur ductile hai.

Aluminium: bartan aur foil, kyunke wo halka aur malleable hai.

Loha aur steel: imaraton ka dhancha, kyunke wo mazboot hai.

Yani har istemal ke peeche ek khasoosiyat hai, aur har khasoosiyat ke peeche wohi ek tasveer: electron ka samundar.

Quiz

1. Describe the **electron sea model** of metallic bonding, and define a **delocalised** electron.
2. Using that model, explain why metals **conduct electricity** and why they are **malleable**. Then give one industrial use each for copper, aluminium and iron, with the property behind it.